

Milestone-Aware On-Policy Distillation for Long-Horizon Reasoning

Anonymous submission

Abstract

On-policy distillation (OPD) can strengthen long-horizon reasoning in compact language models, but it remains unclear which states along a student-generated trajectory warrant teacher supervision. We study how to allocate a limited teacher budget to the states with the greatest influence on downstream recovery. A counterfactual recovery study shows that valuable corrections cluster around semantic stalls and are poorly identified by local disagreement or step-error signals. We propose Milestone-Aware On-Policy Distillation (MA-OPD), which organizes verified teacher solutions into a frozen geometry of semantic milestones and applies the standard OPD loss only to the earliest short region where student progress stalls. MA-OPD requires neither milestone labels nor a learned reward model and adds no test-time machinery. Across six mathematical-reasoning benchmarks, MA-OPD improves mean Avg@8 by 2.15 points over the strongest competing OPD method and by 4.36 points over full-trajectory OPD while requiring $0.41\times$ the teacher-side computation of Full-OPD. It also reaches 95.2% success on ALFWorld. The results show that allocating supervision by milestone progress can improve both performance and teacher efficiency in reasoning and sequential decision-making tasks.

1 Introduction

Extended reasoning trajectories enable language models to solve mathematical problems and make sequential decisions in interactive environments. To deploy these capabilities in compact models, we typically distill a student from a stronger teacher (Hinton, Vinyals, and Dean 2015). Long derivations make this difficult: the student increasingly visits prefixes absent from teacher-generated solutions, and one unproductive segment can derail the remainder of the trajectory. Sequence-level distillation transfers successful teacher generations (Kim and Rush 2016; Lin et al. 2020). On-policy distillation (OPD) reduces the resulting state-distribution mismatch by querying the teacher on student-generated prefixes (Agarwal et al. 2024). Reverse-KL and adaptive sampling variants further improve generative distillation (Gu et al. 2024; Ko et al. 2024).

Yet covering the student’s state distribution does not reveal which states merit supervision. Full-trajectory OPD queries the teacher at every visited state, treating all positions as equally eligible even though teacher cost grows with trajectory length and useful corrective information may be highly

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

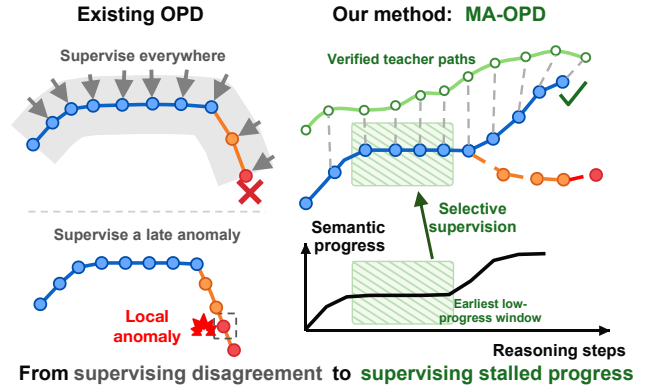


Figure 1: MA-OPD targets the earliest stalled-progress window; existing selectors supervise broadly or react to late anomalies.

localized (Li et al. 2026). Recent methods improve efficiency or reliability in complementary ways. Prefix-OPD prioritizes early positions (Zhang et al. 2026a), whereas TOPD and POPD regulate the rollout horizon (Zhang et al. 2026b). Prune-OPD removes suffixes according to local teacher-student compatibility (Yang et al. 2026); EOPD adapts the divergence at uncertain states (Jin et al. 2026); and RE-OPOLD and TrOPD stabilize learning under policy mismatch (Ko et al. 2026; Xing et al. 2026). These advances estimate whether a teacher target is timely, confident, or locally compatible, but not whether supervision at that state materially improves the solvability of the remaining problem. In long-horizon reasoning, large local disagreement can be a downstream symptom of an earlier failure to advance. Conversely, a state with modest disagreement may mark the point beyond which recovery becomes unlikely.

Intermediate supervision explicitly represents structure within a trajectory. Process reward models assess the correctness of individual reasoning steps (Lightman et al. 2023; Wang et al. 2024), and self-correction methods train models to revise flawed responses (Kumar et al. 2025). In sequential decision-making, subgoals and environment milestones provide denser credit over long horizons (Hashemzadeh et al. 2024; Wang et al. 2026). Such methods demonstrate the value of intermediate structure, but typically depend on step labels,

explicit milestones, a learned reward model, or a task-specific control interface. OPD still lacks a general training-time selector for consequential states on a student’s own trajectory that preserves the distillation objective and requires no task-specific intermediate supervision.

We study a central question: *where should teacher supervision be allocated on an on-policy trajectory to maximize downstream recoverability?* Fig. 1 summarizes the contrast: MA-OPD moves from supervising disagreement to supervising stalled progress. A counterfactual diagnostic study shows that recovery value concentrates near semantic stalls and is poorly captured by local discrepancy or error signals. We then introduce Milestone-Aware On-Policy Distillation (MA-OPD). MA-OPD represents successful teacher reasoning as paths through ordered semantic milestones. It globally matches each student rollout to one verified path and applies the unchanged OPD loss to the earliest short interval with the least progress along that path. This formulation provides milestone-aware credit assignment without explicit milestone annotations or any test-time component.

Our contributions are threefold:

- We identify a supervision-allocation insight for long-horizon OPD: recovery-relevant states cluster around semantic stalls; positions with the largest local disagreement or first visible errors often occur elsewhere.
- We introduce milestone-aware OPD, which uses successful teacher trajectories as a semantic progress reference and identifies consequential regions across alternative solution paths without milestone text, a process reward model, or additional inference machinery.
- Across six mathematical-reasoning benchmarks, MA-OPD improves mean Avg@8 by 2.15 points over the strongest competing OPD method and by 4.36 points over Full-OPD, while requiring $0.41\times$ the teacher-side computation of Full-OPD. The gains persist across model scales and families and transfer to long-horizon embodied decision-making in ALFWorld.

2 Related Work

On-policy knowledge distillation. Classical knowledge distillation matches teacher outputs (Hinton, Vinyals, and Dean 2015); sequence-level and autoregressive variants extend it to trajectories (Kim and Rush 2016; Lin et al. 2020). GKD queries the teacher on student generations to reduce state mismatch (Agarwal et al. 2024). MiniLLM uses reverse KL, and DistiLLM uses adaptive skew divergences (Gu et al. 2024; Ko et al. 2024); other variants add multi-step returns, privileged self-distillation, or peer context (Fan et al. 2026; Zhao et al. 2026; Yu et al. 2026). These methods train on student-visited states but do not select the most consequential positions. MA-OPD retains reverse KL and selects positions by downstream recoverability.

Selective supervision in long-horizon OPD. To reduce full-rollout cost or unreliability (Li et al. 2026), Prefix-OPD and Early Stopping Rollout limit distillation to early prefixes (Zhang et al. 2026a; Zhou et al. 2026), TOPD/POPD control the horizon (Zhang et al. 2026b), and Prune-OPD truncates

suffixes using local top- k compatibility (Yang et al. 2026). EOPD changes the divergence at high entropy (Jin et al. 2026); REOPOLD, TrOPD, and TOP-D stabilize supervision under distribution mismatch (Ko et al. 2026; Xing et al. 2026; Xie et al. 2026). These methods use chronological or local distributional signals that can miss high-recovery bottlenecks. MA-OPD measures semantic progress along verified solution paths while keeping the OPD loss fixed.

Intermediate-state supervision. Process supervision labels intermediate steps (Lightman et al. 2023; Wang et al. 2024); SCoRe trains self-correction with online RL (Kumar et al. 2025); and CoT-Evo refines teacher chains before offline distillation (Feng et al. 2026). Agent methods learn from annotated subgoals (Hashemzadeh et al. 2024) or use environment milestones for reward shaping (Wang et al. 2026). These approaches require step labels, correction rewards, offline trace quality, or explicit environment progress. MA-OPD derives frozen semantic anchors from verified teacher paths, measures student-relative progress, and adds no milestone text, planner, reward model, or test-time component.

3 Preliminary Study

Full-trajectory OPD supervises every state, while position, entropy, reverse KL, and correctness do not measure downstream recoverability. Counterfactual interventions identify high-value supervision states on failed rollouts.

Counterfactual recovery assay. We collect three incorrect initial-student rollouts for each of 180 verifiable mathematics problems (540 trajectories across four domains). After deterministic segmentation, checker-assisted annotators label whether unit j adds a new, correct, and relevant consequence as $a_j \in \{0, 1\}$. The first sustained semantic stall is

$$\tau := \min\{j : a_j = 0 \wedge a_{j+1} = 0\}. \quad (1)$$

At each boundary s_{j-1} , the frozen teacher supplies a verified, answer-free, one-unit bridge c_j ; a length- and style-matched control $\tilde{c}_j$ only restates known information. The unchanged student then samples $K = 16$ continuations from each intervened prefix. Recovery leverage is the difference in verified completion rates:

$$\begin{aligned} \hat{G}_j &:= \frac{1}{K} \sum_{k=1}^K R\left(x, s_{j-1} \circ c_j \circ \tilde{y}_j^{(k)}\right) \\ &\quad - \frac{1}{K} \sum_{k=1}^K R\left(x, s_{j-1} \circ \tilde{c}_j \circ \tilde{y}_{j,\text{ctl}}^{(k)}\right), \end{aligned} \quad (2)$$

where R verifies the final answer. Across 10,024 paired intervention sites, we also record reverse KL, step wrongness, and the first local error.

Findings. Among units at or above the 90th recovery percentile, 70.7% lie within $-2 \leq \Delta j \leq 3$ of the stall boundary. Mean gains in this region are 11.1, 15.7, and 18.9 percentage points for short, medium, and long trajectories. After breaking ties by the earliest recovery maximizer, only 24.6% of KL maxima and 29.3% of first local errors lie within one unit. Across all units, reverse KL and step wrongness

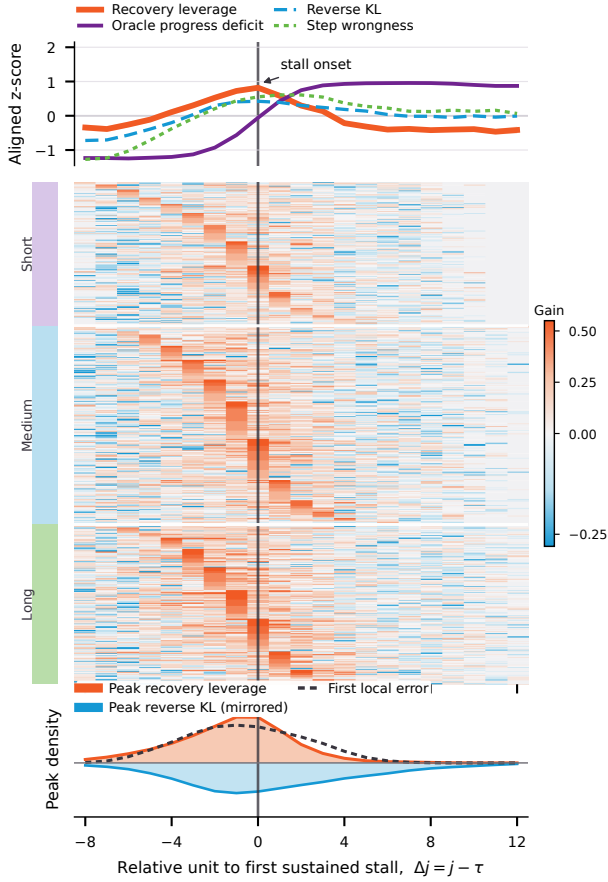


Figure 2: **Stall-aligned recovery cartography.** Recovery leverage for 540 failed trajectories and 10,024 sites aligned by $\Delta j = j - \tau$. Recovery clusters near the stall boundary; reverse-KL peaks and first local errors often occur elsewhere.

correlate weakly with recovery (Spearman $\rho = 0.09$ and 0.11); 42.0% of high-recovery units have at-or-below-median within-trajectory KL.

Implication. The results favor tracking semantic advancement and targeting the earliest short stall window; complete suffixes and local anomalies are less precise. Oracle annotations and bridge interventions are too costly for training, so MA-OPD uses frozen teacher-path geometry to select the earliest minimum-progress window.

4 Method

Milestone-Aware On-Policy Distillation (MA-OPD) represents verified teacher solutions as paths through ordered semantic milestones. A frozen encoder and truncated singular value decomposition (SVD) define a low-dimensional geometry for each prompt (Figure 3). Each completed on-policy rollout is matched to one teacher path; MA-OPD then finds the earliest fixed-span window with the least progress and applies reverse-KL OPD only there. No milestone text, learned progress classifier, or test-time component is required.

Problem Setup

Let $x \sim \mathcal{D}$ be a prompt, π_θ the student, and π_{tea} a frozen teacher with full-vocabulary logits. A verifier returns $R(x, y) \in \{0, 1\}$. At each iteration, the stop-gradient behavior copy $\pi_{\bar{\theta}}$, with $\bar{\theta} \leftarrow \text{sg}(\theta)$, samples $y = (y_1, \dots, y_T) \sim \pi_{\bar{\theta}}(\cdot | x)$. For state $s_t = (x, y_{<t})$, the reverse-KL loss is

$$\ell_{\text{OPD}}(s_t; \theta) := \sum_{v \in \mathcal{V}} \pi_\theta(v | s_t) \log \frac{\pi_\theta(v | s_t)}{\pi_{\text{tea}}(v | s_t)}. \quad (3)$$

Here, $\mathcal{V}$ is the shared vocabulary and the teacher receives the original student prefix. Full-OPD averages Eq. (3) over all tokens; MA-OPD restricts the same loss to selected positions (Agarwal et al. 2024; Gu et al. 2024; Ko et al. 2024).

A deterministic segmenter gives reasoning units $\sigma(y) = (u_1, \dots, u_J)$. If b_j is the last token of u_j , with $b_0 = 0$, then $\bar{s}_j = (x, y_{\leq b_j})$ is its boundary prefix. Progress is measured at boundaries, while distillation remains token-level.

Low-Rank Milestone Geometry

Verified solutions may use different wording while sharing semantic stages. We model them as curves in the frozen initial student’s state space and fit a shared low-rank transition subspace. A milestone is an ordered path anchor, not a manually labeled fact.

Frozen semantic states. Let $H_{\theta_0}^{(\text{last})}(s)$ denote the initial student’s final-layer hidden states on prefix s . We define the fixed representation map

$$g_0(s) := \frac{\text{Pool}(H_{\theta_0}^{(\text{last})}(s))}{\left\| \text{Pool}(H_{\theta_0}^{(\text{last})}(s)) \right\|_2}. \quad (4)$$

The pool averages the latest reasoning unit, or the prompt tokens at the initial state. Copied from π_{θ_0} , g_0 remains frozen during training.

For prompt x , collect $\widetilde{M}_x$ verified teacher paths $\tau_x^r = (u_1^r, \dots, u_{K_r}^r)$, with $s_k^r = (x, u_{\leq k}^r)$, $z_k^r = g_0(s_k^r) \in \mathbb{R}^d$, and shared origin $z_0 = g_0(x)$. Their transition matrix is

$$\Delta z_k^r := z_k^r - z_{k-1}^r, \quad G_x := [\Delta z_1^1, \dots, \Delta z_{K_1}^1, \dots, \Delta z_{K_{\widetilde{M}_x}}^{\widetilde{M}_x}] \in \mathbb{R}^{d \times N_x}, \quad (5)$$

where $N_x = \sum_r K_r$; each column is one verified-path transition.

Spectral progress subspace. The complete spectral construction is

$$\begin{aligned} G_x &= \widetilde{U}_x \Sigma_x V_x^\top, \\ m_x &= \min \left\{ m : \frac{\sum_{i=1}^m \sigma_i^2}{\sum_i \sigma_i^2} \geq \eta \right\}, \quad U_x = \widetilde{U}_x[:, 1 : m_x], \\ \Pi_x &= U_x U_x^\top, \\ a_k^r &= U_x^\top (z_k^r - z_0) \in \mathbb{R}^{m_x}, \quad a_0^r = 0. \end{aligned} \quad (6)$$

With descending σ_i , m_x is the smallest rank retaining an η fraction of transition energy. Thus, U_x is the optimal rank- m_x transition subspace by Eckart–Young, Π_x its projector, and a_k^r the path coordinates.

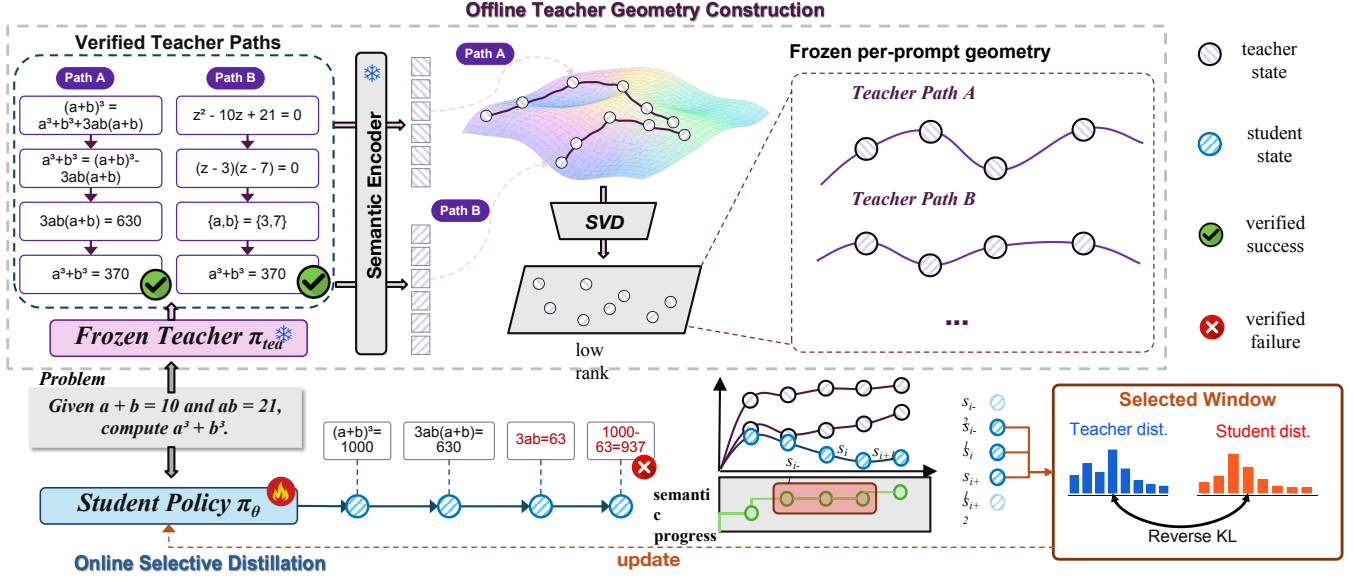


Figure 3: MA-OPD pipeline. Offline, a frozen encoder and SVD convert verified teacher paths into a per-prompt milestone geometry. Online, MA-OPD matches each rollout to one path and distills only its earliest minimum-progress window. The global match assigns every boundary a shared, path-normalized progress coordinate, preventing local switches between candidate paths. Teacher scoring and reverse-KL are confined to the selected window, reducing training cost without changing inference. Only the student is retained after training.

We merge segments with $\|a_k^r - a_{k-1}^r\|_2 \leq \epsilon_{\text{seg}}$, discard empty paths, and reindex the remaining polylines $\mathcal{P}_x = \{(a_0^r, \dots, a_{K_r}^r)\}_{r=1}^{M_x}$, where $M_x \leq \tilde{M}_x$. Both U_x and $\mathcal{P}_x$ are built once and frozen.

Geometric Progress Projection

To obtain one coherent progress coordinate, we match the complete rollout to the verified path with minimum cumulative projection residual and score every boundary against that path. The full-rollout decision prevents adjacent prefixes from switching paths when candidate solutions cross in representation space. Arc-length normalization separates semantic advance from differences in solution granularity. The resulting coordinate is continuous and needs no entailment classifier or milestone-specific threshold.

Student coordinates. For on-policy boundary prefix $\bar{s}_j$ and segment k of teacher path r , define the student coordinate, segment direction, projection fraction, and residual:

$$\begin{aligned} q_j &= U_x^\top (g_0(\bar{s}_j) - z_0), & d_k^r &= a_k^r - a_{k-1}^r, \\ \alpha_k^r(j) &:= \text{clip}_{[0,1]} \frac{\langle d_k^r, q_j - a_{k-1}^r \rangle}{\|d_k^r\|_2^2}, \\ e_k^r(j) &:= \|q_j - a_{k-1}^r - \alpha_k^r(j)d_k^r\|_2^2. \end{aligned} \quad (7)$$

Here, $\alpha_k^r(j)$ locates the closest point on segment d_k^r , and $e_k^r(j)$ is its squared residual. Let $A_k^r = \sum_{\nu=1}^k \|d_\nu^r\|_2$, with $A_0^r = 0$, denote cumulative arc length. Global path matching

and normalized progress are

$$\begin{aligned} \varepsilon_j^r &:= \min_{1 \leq k \leq K_r} e_k^r(j), & E^r(x, y) &:= \sum_{i=0}^J \varepsilon_i^r, \\ \hat{r} &:= \min \arg \min_{1 \leq r \leq M_x} E^r(x, y), \\ \hat{k}_j &:= \min \arg \min_{1 \leq k \leq K_{\hat{r}}} e_k^{\hat{r}}(j), \\ \rho_j &:= \frac{A_{\hat{k}_j-1}^{\hat{r}} + \alpha_{\hat{k}_j}^{\hat{r}}(j) \|d_{\hat{k}_j}^{\hat{r}}\|_2}{A_{K_{\hat{r}}}^{\hat{r}}} \in [0, 1], \\ \bar{\rho}_j &:= \max_{0 \leq i \leq j} \rho_i, & \bar{\rho}_0 &= 0. \end{aligned} \quad (8)$$

The complete rollout fixes $\hat{r}$; $\hat{k}_j$ locates each boundary on it. A shared arc-length denominator prevents jumps between paths. Ties choose the smaller path index and then the earlier segment, while $\bar{\rho}_j$ preserves previously attained progress.

The cumulative residual E^r chooses the path; window selection uses changes in $\bar{\rho}_j$. We retain $\varepsilon_j = \varepsilon_j^{\hat{r}}$ only as a diagnostic. The default omits scale-dependent residual gating, which we evaluate through stratified results and a quantile-gated ablation.

Minimum-Progress Window Localization

To target the earliest bottleneck and avoid supervising downstream errors, we select the earliest fixed-span interval with minimum geometric progress.

Window selection. For a horizon of H reasoning units, selection is valid when $\forall(x, y)$ holds: $R(x, y) = 0, J \geq H$,

Method	LLM pair: student $\leftarrow$ teacher	Avg@8						Mean	Pass avg.	Long-Q4	
		MATH-500	AMC23	AIME24	AIME25	Minerva	Olympiad				
■ <i>Unified objective and horizon baselines</i>											
KD (Kim and Rush 2016)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	68.4	42.7	11.6	7.4	31.1	31.8	32.17	48.7	22.9	
Full-OPD (Agarwal et al. 2024)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	72.9	47.8	16.4	10.7	34.8	35.9	36.42	55.8	29.8	
Prefix-OPD (Zhang et al. 2026a)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	72.3	47.1	15.7	10.2	34.4	35.2	35.82	54.9	28.6	
TOPD (Zhang et al. 2026b)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	72.7	47.6	16.0	10.4	34.6	35.6	36.15	55.4	29.2	
POPD (Zhang et al. 2026b)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	73.3	48.1	16.8	10.9	35.0	36.1	36.70	56.2	30.4	
Prune-OPD (Yang et al. 2026)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	73.7	48.6	17.2	11.3	35.3	36.4	37.08	56.9	31.0	
EOPD (Jin et al. 2026)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	74.2	49.3	18.1	11.9	35.8	37.0	37.72	58.5	31.8	
REOPOLD (Ko et al. 2026)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	74.7	49.8	18.5	12.4	36.1	37.3	38.13	59.2	32.4	
TrOPD (Xing et al. 2026)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	<u>75.2</u>	<u>50.4</u>	<u>19.1</u>	<u>12.8</u>	<u>36.5</u>	<u>37.8</u>	<u>38.63</u>	<u>60.1</u>	<u>33.1</u>	
■ <i>Equal-span supervision selectors</i>											
Position	Qwen3-1.7B $\leftarrow$ Qwen3-8B	71.8	46.9	15.2	9.8	34.0	34.9	35.43	54.1	27.8	
Random	Qwen3-1.7B $\leftarrow$ Qwen3-8B	72.4	47.3	15.8	10.1	34.3	35.3	35.87	54.8	28.5	
Teacher entropy	Qwen3-1.7B $\leftarrow$ Qwen3-8B	73.5	48.5	17.1	11.2	35.2	36.3	36.97	56.7	30.7	
Reverse-KL	Qwen3-1.7B $\leftarrow$ Qwen3-8B	73.8	48.8	17.4	11.5	35.4	36.6	37.25	57.1	30.9	
Top- k drift	Qwen3-1.7B $\leftarrow$ Qwen3-8B	74.0	49.0	17.7	11.6	35.6	36.8	37.45	57.5	31.3	
Step error	Qwen3-1.7B $\leftarrow$ Qwen3-8B	74.4	49.5	18.0	12.0	35.9	37.1	37.82	58.0	31.8	
■ <i>Spectral progress geometry</i>											
MA-OPD (ours)	Qwen3-1.7B $\leftarrow$ Qwen3-8B	76.8 _{+1.6}	52.3 _{+1.9}	22.0 _{+2.9}	15.1 _{+2.3}	38.4 _{+1.9}	40.1 _{+2.3}	40.78 _{+2.15}	64.3 _{+4.2}	37.0 _{+3.9}	
■ <i>Backbone transfer of MA-OPD</i>											
MA-OPD (ours)	Qwen3-4B $\leftarrow$ Qwen3-8B	82.4	64.1	23.6	21.4	42.2	46.3	46.67	67.9	47.5	
MA-OPD (ours)	DS-R1-Qwen-1.5B $\leftarrow$ DS-R1-Qwen-7B	89.1	82.1	42.6	35.4	40.5	59.0	58.12	75.8	55.7	

Table 1: Main mathematical-reasoning results (%). Benchmark columns are Avg@8; Mean and Pass avg. average six benchmarks; Long-Q4 is the longest horizon quartile. The primary block shares models, prompts, rollouts, and seeds. Bold/underline mark best/second; deltas are gains over TrOPD.

and $\mathcal{P}_x \neq \emptyset$. The selected interval is

$$\begin{aligned}
\Delta_H(j) &:= \bar{\rho}_{j+H} - \bar{\rho}_j, & \mathcal{J}_H(y) &= \{0, \dots, J - H\}, \\
j^* &:= \min \arg \min_{j \in \mathcal{J}_H(y)} \Delta_H(j), & \text{when } \mathcal{V}(x, y), \\
W(x, y) &:= \begin{cases} \{b_{j^*} + 1, \dots, b_{j^*+H}\}, & \mathcal{V}(x, y), \\ \emptyset, & \text{otherwise.} \end{cases}
\end{aligned} \tag{9}$$

Thus, $\Delta_H(j) \geq 0$ measures progress across H units; ties select the earliest interval. The span H is the only structural localization choice, and each rollout contributes at most one window.

Selective On-Policy Distillation

MA-OPD changes only the loss support: teacher targets, reverse KL, and the optimizer match Full-OPD, but teacher scoring covers only the selected window.

Training objective. Let $W = W(x, y)$, $\ell_t(\theta) := \ell_{\text{OPD}}(s_t; \theta)$, and let $\mathbb{E}_{\bar{\theta}}$ average over $x \sim \mathcal{D}$ and $y \sim \pi_{\bar{\theta}}(\cdot | x)$:

$$\mathcal{L}_W(x, y; \theta) := \begin{cases} |W|^{-1} \sum_{t \in W} \ell_t(\theta), & W \neq \emptyset, \\ 0, & W = \emptyset, \end{cases} \tag{10}$$

$$\begin{aligned}
\mathcal{L}_{\text{MA-OPD}}(\theta) &:= \mathbb{E}_{\bar{\theta}} [\mathcal{L}_W(x, y; \theta)], \\
\mathcal{L}_{\text{train}}(\theta) &:= \mathcal{L}_{\text{MA-OPD}}(\theta) + \lambda_{\text{anc}} \mathcal{L}_{\text{anc}}(\theta).
\end{aligned}$$

The expectation includes $W = \emptyset$ and uses the full rollout-batch denominator, preserving the selection rate and Full-OPD normalization. The teacher scores original prefixes only at positions in W ; neither geometry nor milestone text enters its context.

The anchor is token-level negative log-likelihood on a fixed supervised batch, with the same λ_{anc} for all OPD baselines. It always uses the full anchor batch, including when no rollout window is selected.

5 Experiments

We organize the experiments around three questions. **Q1:** Does MA-OPD improve task performance, especially on long reasoning horizons, and transfer across model families and domains? **Q2:** Does milestone progress locate recovery-relevant windows while improving the accuracy–compute trade-off? **Q3:** Which design choices matter, and do the training dynamics support the proposed mechanism?

Experimental Setup

Benchmarks and metrics. Mathematical benchmarks comprise MATH-500 (Lightman et al. 2023), AMC 2023, AIME 2024/2025, Minerva Math (Lewkowycz et al. 2022), and OlympiadBench-Math-En (He et al. 2024). We report benchmark-wise Avg@8, mean Pass@8, and Long-Q4 accuracy. ALFWorld (Shridhar et al. 2021), an external long-horizon agent setting, is evaluated by overall and length-stratified success. Additional measures are selector localization and total training cost.

Baselines and fairness. Baselines are standard KD, Full-OPD (Agarwal et al. 2024), and recent efficient OPD variants (Zhang et al. 2026a,b; Yang et al. 2026; Jin et al. 2026; Ko et al. 2026; Xing et al. 2026). Equal-span position, random, entropy, reverse-KL, top- k drift, and step-error selectors isolate milestone geometry; all share rollouts, loss, selected-token, and teacher-scoring budgets. We report fixed-update and compute-matched (teacher/end-to-end) results, keeping

Method	Short	Medium	Long	Overall
■ <i>Non-distillation agent baselines</i>				
Direct prompting	0.0	0.0	0.0	0.0
PPO (Schulman et al. 2017)	30.6	28.4	18.7	26.5
ReAct (Yao et al. 2023)	18.2	10.5	2.0	12.8
GRPO (Shao et al. 2024)	76.7	73.9	53.5	72.8
RLOO (Ahmadian et al. 2024)	55.3	46.3	38.2	47.7
GiGPO (Feng et al. 2025)	90.7	84.3	79.5	86.1
BEACON (Wang et al. 2026)	96.8	87.0	92.9	91.4
■ <i>On-policy distillation</i>				
Full-OPD (Agarwal et al. 2024)	89.5	82.8	74.9	84.2
MA-OPD-Env	96.6	91.8	92.4	93.7
MA-OPD	97.3 _{+0.5}	93.4 _{+6.4}	94.8 _{+1.9}	95.2 _{+3.8}

Table 2: ALFWorld success (%) with Qwen2.5-1.5B; distillation uses a Qwen2.5-7B teacher. ReAct, GRPO, GiGPO, and BEACON are paper-reported; the others use our protocol. Deltas are against the strongest non-MA-OPD method.

methods with different supervision or inference assumptions separate.

Backbones and implementation. The primary setup pairs a Qwen3-1.7B-Base student trained on deduplicated DAPO-Math-17K (Yu et al. 2025) with a frozen non-thinking Qwen3-8B teacher (Qwen Team 2025). Defaults are at most two verified paths, $\eta = 0.90$ spectral energy, and $H = 3$. Each rollout matches the teacher path with the smallest cumulative residual; geometry is built once from the initial student and frozen. Mathematical experiments use three seeds with paired evaluation randomness. Appendix B contains full protocols and implementation details.

Q1: Task Performance and Generalization

Mathematical reasoning. Under the unified protocol (Table 1), MA-OPD reaches 40.78 mean Avg@8, 4.36 points above Full-OPD and 2.15 above TrOPD, with larger AIME and Long-Q4 gains. Improvements over TrOPD remain significant on all six benchmarks after Holm correction and transfer across Qwen3 and DeepSeek-R1-Distill students and native protocols (Appendix C).

Long-horizon agents. On ALFWorld (Table 2), MA-OPD reaches 95.2% overall and 94.8% on Long, beating Full-OPD by 19.9 points and BEACON by 1.9 on Long. MA-OPD-Env performs similarly. A unified three-seed rerun yields gains over BEACON of 3.6 overall and 2.1 on Long ($p = 0.002, 0.018$ after Holm correction; Appendix C).

Q1 takeaway. MA-OPD improves all six reasoning benchmarks and ALFWorld, transfers across backbones, and gains most on AIME and Long-Q4.

Q2: Selector Quality and Efficiency

Localization and cost. MA-OPD attains 68.7% Loc@1 versus 34.1% for the strongest equal-span selector, at $0.41\times$ teacher and $0.79\times$ end-to-end cost (Table 3). The latter also includes student rollouts and projection, so teacher scoring

Method	Loc@1 $\uparrow$	Long-Q4 $\uparrow$	Teacher $\downarrow$	E2E $\downarrow$
■ <i>Trajectory and horizon baselines</i>				
Full-OPD (Agarwal et al. 2024)	—	29.8	1.00	1.00
Prefix-OPD (Zhang et al. 2026a)	21.8	28.6	0.24	0.33
TOPD (Zhang et al. 2026b)	23.7	29.2	0.53	0.58
POPD (Zhang et al. 2026b)	26.2	30.4	0.36	0.44
Prune-OPD (Yang et al. 2026)	33.5	31.0	0.61	0.63
■ <i>Equal-span supervision selectors</i>				
Teacher entropy	27.8	30.7	0.37	0.77
Reverse-KL	24.6	30.9	0.37	0.78
Top- k drift	34.1	31.3	0.38	0.79
Step error	29.3	31.8	0.39	0.86
■ <i>Spectral progress geometry</i>				
MA-OPD	68.7 _{+34.6}	37.0 _{+5.2}	0.41	0.79

Table 3: Localization, Long-Q4, and cost for Qwen3-1.7B $\leftarrow$ Qwen3-8B. Loc@1 is agreement within one oracle unit; costs are relative to Full-OPD.

Method	Fixed updates	Teacher matched	E2E matched
■ <i>Trajectory and horizon baselines</i>			
Full-OPD (Agarwal et al. 2024)	36.42 / 29.8	34.88 / 27.5	35.96 / 29.0
Prefix-OPD (Zhang et al. 2026a)	35.82 / 28.6	36.36 / 29.2	36.51 / 29.6
TOPD (Zhang et al. 2026b)	36.15 / 29.2	35.62 / 28.4	36.41 / 29.5
POPD (Zhang et al. 2026b)	36.70 / 30.4	37.11 / 30.8	37.08 / 30.8
Prune-OPD (Yang et al. 2026)	37.08 / 31.0	36.34 / 29.9	37.19 / 31.2
■ <i>Objective and trajectory baselines</i>			
EOPD (Jin et al. 2026)	37.72 / 31.8	35.74 / 29.3	37.39 / 31.4
REOPOLD (Ko et al. 2026)	38.13 / 32.4	36.11 / 30.0	37.80 / 32.0
TrOPD (Xing et al. 2026)	<u>38.63</u> / <u>33.1</u>	36.72 / <u>31.0</u>	<u>38.04</u> / <u>32.4</u>
■ <i>Spectral progress geometry</i>			
MA-OPD (ours)	40.78 / 37.0	40.78 / 37.0	40.78 / 37.0

Table 4: Compute-matched Mean Avg@8 / Long-Q4 at 33.8 teacher and 104.8 end-to-end A100 GPU-hours; bold/underline mark best/strongest baseline.

provides most savings. Thus, the 34.6-point Loc@1 advantage over Top- k drift comes at essentially the same total cost.

Compute-matched performance. MA-OPD leads by 3.67 mean Avg@8 and 6.0 Long-Q4 points at equal teacher compute, and by 2.74 and 4.6 over TrOPD at equal end-to-end cost (Table 4). The same 40.78/37.0 checkpoint satisfies both caps, so the gain cannot be explained by extra teacher scoring or longer training.

Recovery-oracle agreement. MA-OPD never observes the recovery oracle during training; it is built only for diagnosis on 540 failed trajectories from 180 problems. At each boundary, a verified one-unit teacher bridge is inserted and the student samples 16 continuations. Recovery leverage is the completion-rate gain over a restatement control; the oracle is the earliest boundary with the largest gain. For six $H = 3$ selectors, we report proximity (Loc@ d /Hit@5), distance (error), direction (bias), and missed gain (regret; Appendix B). Agreement asks whether supervision falls where a small intervention most improves continuation. On this diagnostic, MA-OPD reaches 45.2/68.7/82.4% Loc@0/1/2, versus 34.1/48.9% Loc@1/2 for Top- k Drift (Figure 5). Error/regret fall from 2.73/5.59 to 1.32/2.46, and bias is +0.15

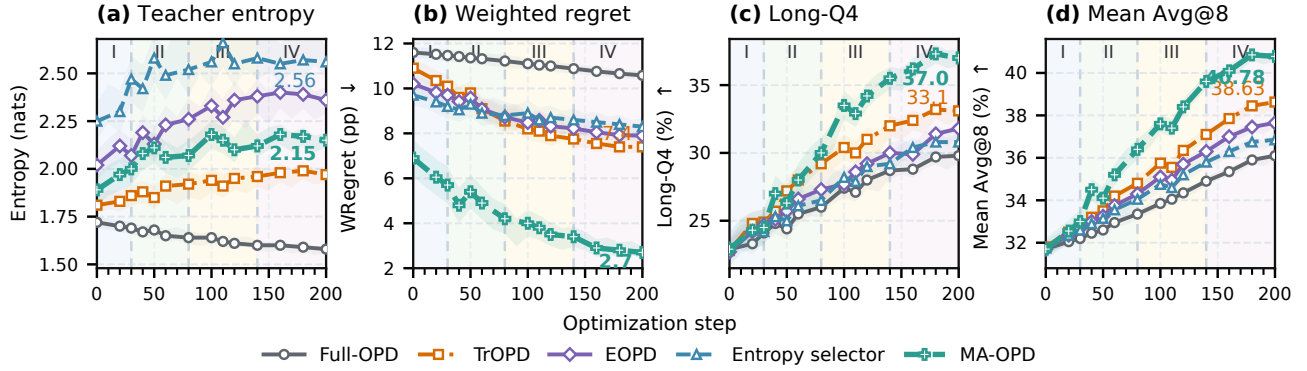


Figure 4: Training dynamics: (a) teacher entropy, (b) WRegret, (c) Long-Q4, and (d) Mean Avg@8 (mean ± 1 standard deviation, three runs). Shading marks the four optimization phases.

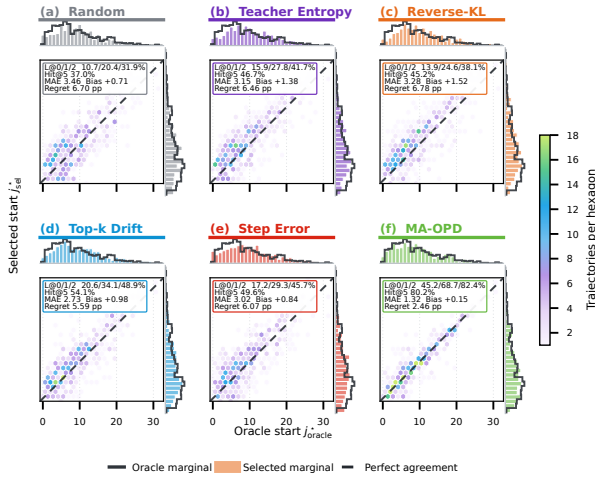


Figure 5: Oracle versus selected starts on 540 failed trajectories. Diagonals show exact agreement; marginals and insets summarize localization and regret.

versus $+1.38/+1.52$ for entropy/reverse-KL. The near-zero bias also shows that MA-OPD does not systematically shift supervision later in the trajectory. This supports, but does not prove, the mechanism. Although exact agreement is 45.2%, 82.4% of selections fall within two units, so most residual errors are local instead of trajectory-scale misses.

Q2 takeaway. Milestone progress better matches recoverability, and its gains persist at matched teacher and end-to-end cost without greater teacher use.

Q3: Design and Mechanism

Structural ablations. Transition SVD improves Avg@8/Loc@1 by 0.91/9.3 points (Appendix Figure 13); two paths add 0.85/6.2, while a third adds little. Refreshing raises cost from $0.79\times$ to $0.96\times$; gates trade 0.15–0.34

Avg@8 for 0.9–1.4 Loc@1. Frozen, ungated two-path geometry is best; matched controls appear in Appendix C. This favors a compact, fixed state-change representation. The first two paths capture most of the useful solution diversity in this setting. Freezing the geometry also avoids rebuilding the state representation and path coordinates as the student changes.

Training dynamics. Across 200 steps, methods share the training and evaluation protocol; three runs use paired randomness. We track Long-Q4, Mean Avg@8, WRegret (Appendix B), and teacher entropy at supervised states. At step 200, MA-OPD targets lower-entropy states than the entropy selector (2.15 vs. 2.56 nats) and has WRegret 2.7, versus 7.4 for TrOPD and 10.58 for Full-OPD (Figure 4(a–b)). Its frozen selector shows that these changes arise from evolving on-policy states. WRegret separates before the accuracy curves and precedes the Long-Q4 gap (Figure 4(c–d)), making localization quality a leading signal, not a post hoc explanation. At step 200, MA-OPD reaches 37.0/40.78 Long-Q4/mean Avg@8, versus 33.1/38.63 for TrOPD and 29.8/36.10 for Full-OPD. Its 3.9-point Long-Q4 margin over TrOPD is nearly twice the 2.15-point mean Avg@8 margin. The advantage grows with horizon instead of shifting scores uniformly, consistent with early stalls compounding across downstream steps.

Q3 takeaway. Supervision location matters most: frozen two-path geometry lowers WRegret ahead of Long-Q4 gains; extra paths, refresh, and gradient matching provide little benefit.

6 Conclusion

MA-OPD targets the earliest semantic stall with reverse-KL and no test-time overhead. Mathematics and ALFWorld gains with less teacher scoring support semantic progress for efficient long-horizon distillation. Compute-matched ablations identify supervision location—not teacher use or update size—as the main driver. Future work will quantify how many noisy teacher paths suffice for stable geometry.

References

- Agarwal, R.; Vieillard, N.; Zhou, Y.; Stanczyk, P.; Ramos Garea, S.; Geist, M.; and Bachem, O. 2024. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. In *International Conference on Learning Representations*.
- Ahmadian, A.; Cremer, C.; Gallé, M.; Fadaee, M.; Kreutzer, J.; Pietquin, O.; Üstün, A.; and Hooker, S. 2024. Back to Basics: Revisiting REINFORCE-Style Optimization for Learning from Human Feedback in LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 12248–12267. Association for Computational Linguistics.
- Fan, J.; Luo, G.; Bowling, M.; and Mou, L. 2026. KETCHUP: K-Step Return Estimation for Sequential Knowledge Distillation. In *Findings of the Association for Computational Linguistics: EACL 2026*, 778–796.
- Feng, K.; Ding, K.; Zhu, Z.; Liang, L.; Zhang, Q.; and Chen, H. 2026. CoT-Evo: Evolutionary Distillation of Chain-of-Thought for Scientific Reasoning. In *International Conference on Learning Representations*.
- Feng, L.; Xue, Z.; Liu, T.; and An, B. 2025. Group-in-Group Policy Optimization for LLM Agent Training. In *Advances in Neural Information Processing Systems*, volume 38.
- Gu, Y.; Dong, L.; Wei, F.; and Huang, M. 2024. MiniLLM: Knowledge Distillation of Large Language Models. In *International Conference on Learning Representations*.
- Hashemzadeh, M.; Stengel-Eskin, E.; Chandar, S.; and Côté, M.-A. 2024. Sub-goal Distillation: A Method to Improve Small Language Agents. In *Conference on Lifelong Learning Agents*.
- He, C.; Luo, R.; Bai, Y.; Hu, S.; Thai, Z.; Shen, J.; Hu, J.; Han, X.; Huang, Y.; Zhang, Y.; Liu, J.; Qi, L.; Liu, Z.; and Sun, M. 2024. OlympiadBench: A Challenging Benchmark for Promoting AGI with Olympiad-Level Bilingual Multimodal Scientific Problems. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 3828–3850.
- Hendrycks, D.; Burns, C.; Kadavath, S.; Arora, A.; Basart, S.; Tang, E.; Song, D.; and Steinhardt, J. 2021. Measuring Mathematical Problem Solving With the MATH Dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1.
- Hinton, G.; Vinyals, O.; and Dean, J. 2015. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.
- Jin, W.; Min, T.; Yang, Y.; Wei, D.; Zhou, Y.; Kadhe, S. R.; Baracaldo, N.; and Lee, K. 2026. Entropy-Aware On-Policy Distillation of Language Models. In *Proceedings of the 43rd International Conference on Machine Learning*.
- Kim, Y.; and Rush, A. M. 2016. Sequence-Level Knowledge Distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, 1317–1327.
- Ko, J.; Abdali, S.; Kim, Y. J.; Chen, T.; and Cameron, P. 2026. Scaling Reasoning Efficiently via Relaxed On-Policy Distillation. arXiv:2603.11137.
- Ko, J.; Kim, S.; Chen, T.; and Yun, S.-Y. 2024. DistiLLM: Towards Streamlined Distillation for Large Language Models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, 24872–24895.
- Kumar, A.; Zhuang, V.; Agarwal, R.; Su, Y.; Co-Reyes, J. D.; Singh, A.; Baumli, K.; Iqbal, S.; Bishop, C.; Roelofs, R.; Zhang, L. M.; Kinney, K.; Shrivastava, D.; Paduraru, C.; Tucker, G.; Precup, D.; Behbahani, F.; and Faust, A. 2025. Training Language Models to Self-Correct via Reinforcement Learning. In *International Conference on Learning Representations*.
- Lewkowycz, A.; Andreassen, A.; Dohan, D.; Dyer, E.; Michalewski, H.; Ramasesh, V.; Slone, A.; Anil, C.; Schlag, I.; Gutman-Solo, T.; Wu, Y.; Neyshabur, B.; Gur-Ari, G.; and Misra, V. 2022. Solving Quantitative Reasoning Problems with Language Models. In *Advances in Neural Information Processing Systems*, volume 35.
- Li, Y.; Zuo, Y.; He, B.; Zhang, J.; Xiao, C.; Qian, C.; Yu, T.; Gao, H.-a.; Yang, W.; Liu, Z.; and Ding, N. 2026. Rethinking On-Policy Distillation of Large Language Models: Phenomenology, Mechanism, and Recipe. arXiv:2604.13016.
- Lightman, H.; Kosaraju, V.; Burda, Y.; Edwards, H.; Baker, B.; Lee, T.; Leike, J.; Schulman, J.; Sutskever, I.; and Cobbe, K. 2023. Let’s Verify Step by Step. arXiv:2305.20050.
- Lin, A.; Wohlwend, J.; Chen, H.; and Lei, T. 2020. Autoregressive Knowledge Distillation through Imitation Learning. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 6121–6133.
- Qwen Team. 2025. Qwen3 Technical Report. arXiv:2505.09388.
- Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal Policy Optimization Algorithms. arXiv:1707.06347.
- Shao, Z.; Wang, P.; Zhu, Q.; Xu, R.; Song, J.; Bi, X.; Zhang, H.; Zhang, M.; Li, Y. K.; Wu, Y.; and Guo, D. 2024. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. arXiv:2402.03300.
- Shridhar, M.; Yuan, X.; Côté, M.-A.; Bisk, Y.; Trischler, A.; and Hausknecht, M. 2021. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. In *International Conference on Learning Representations*.
- Wang, P.; Li, L.; Shao, Z.; Xu, R.; Dai, D.; Li, Y.; Chen, D.; Wu, Y.; and Sui, Z. 2024. Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 9426–9439.
- Wang, Z.; Yan, Y.; Li, H.; Pan, T.; Li, D.; Zhang, R.; Lu, W.; Xiao, J.; Zhuang, Y.; and Shen, Y. 2026. Milestone-Guided Policy Learning for Long-Horizon Language Agents. arXiv:2605.06078.
- Xie, Z.; Zhang, L. L.; Xie, Z.; and Yang, M. 2026. Trust Region Policy Distillation. arXiv:2607.04751.
- Xing, X.; Wang, H.; Gao, B.; Li, Z.; and Tang, Y. 2026. Trust Region On-Policy Distillation. arXiv:2606.01249.

Yang, Z.; Guo, Z.; Song, Y.; Xu, M.; Wang, Y.; Wang, Y.; Liang, X.; and Tang, J. 2026. Prune-OPD: Efficient and Reliable On-Policy Distillation for Long-Horizon Reasoning. arXiv:2605.07804.

Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*.

Yu, Q.; Zhang, Z.; Zhu, R.; Yuan, Y.; Zuo, X.; Yue, Y.; Dai, W.; Fan, T.; Liu, G.; Liu, J.; Liu, L.; Liu, X.; Lin, H.; Lin, Z.; Ma, B.; Sheng, G.; Tong, Y.; Zhang, C.; Zhang, M.; Zhang, R.; Zhang, W.; Zhu, H.; Zhu, J.; Chen, J.; Chen, J.; Wang, C.; Yu, H.; Song, Y.; Wei, X.; Zhou, H.; Liu, J.; Ma, W.-Y.; Zhang, Y.-Q.; Yan, L.; Wu, Y.; and Wang, M. 2025. DAPO: An Open-Source LLM Reinforcement Learning System at Scale. In *Advances in Neural Information Processing Systems*, volume 38.

Yu, W.; Li, X.; Zhao, Y.; Liu, X.; Zhang, R.; Wang, H.; Luo, Y.; Wu, C. H.; Mittal, G.; Fredrikson, M.; and Hu, Y. 2026. Multi-Rollout On-Policy Distillation via Peer Successes and Failures. arXiv:2605.12652.

Zhang, D.; Yang, Z.; Janghorbani, S.; Han, J.; Ressler II, A.; Qian, Q.; Lyng, G. D.; Batra, S. S.; and Tillman, R. E. 2026a. Fast and Effective On-Policy Distillation from Reasoning Prefixes. In *Findings of the Association for Computational Linguistics: ACL 2026*, 25553–25569. Association for Computational Linguistics.

Zhang, Y.; Chai, J.; Tu, S.; Fu, Y.; Wang, X.; Lin, W.; Yin, G.; Zhang, Q.; Zhu, Y.; and Zhao, D. 2026b. Are Full Rollouts Necessary for On-Policy Distillation? arXiv:2605.31490.

Zhao, S.; Xie, Z.; Liu, M.; Huang, J.; Pang, G.; Chen, F.; and Grover, A. 2026. Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models. arXiv:2601.18734.

Zhou, Z.; Li, J.; Tang, H.; Wu, Y. N.; and Terzopoulos, D. 2026. Less is More: Early Stopping Rollout for On-Policy Distillation. arXiv:2605.27028.